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Marine Structures 20 (2007) 164–184
MARINE STRUCTURES
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Wave-in-deck loading on fixed steel jacket decks

Katrine van Raaij[a,*], Ove T. Gudmestad[a,b]

[a]*University of Stavanger, Stavanger, Norway*
[b]*Statoil, Stavanger, Norway*

Received 30 March 2005; accepted 30 March 2007

Abstract

For quite some years, wave-in-deck loading has been an issue of concern for engineers dealing with the performance of offshore structures. The topic is particularly relevant for reassessment of existing structures located in subsiding areas. One agrees that wave-in-deck loading is of dynamic nature, and that structural analyses should reflect this. There is, however, no general consensus on the size of the load and the shape of the load time history to be used for such analyses.

In this paper focus has been on finding realistic load time histories for wave-in-deck loading on jacket platforms in the North Sea. A (normalised) time history shape and a simple expression to calculate a reference load (maximum load) to quantify the time history is presented.

The presented 'recipe' for time histories is based on experimental data and is supported by results reported in the literature, comprising relevant computer simulations and model experiments of wave-in-deck loads on fixed offshore structures.

The recommended load time history is intended for analyses where detailed information on the deck load for a given structure is unavailable, and where a simplified 'rough-but-reasonable' estimate can be accepted.
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Keywords: Wave-in-deck; Dynamic analysis; Time domain analysis; Pushover analysis; Wave experiments

1. Introduction

Wave-in-deck loads potentially represent treats to offshore platforms in case of wave crests being higher than the crests the platforms have been designed for or in the case of

*Corresponding author.
E-mail addresses: katrine.v.raaij@uis.no (K. van Raaij), otg@statoil.com (O.T. Gudmestad).

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doi:10.1016/j.marstruc.2007.03.003

subsidence of the seafloor caused by hydrocarbon extraction in the ground. The question of wave-in-deck loading has recently gained interest in the aftermath of hurricane damages in the Gulf of Mexico and for subsiding platforms in the North Sea.

(Re-)analysis of offshore structures, in particular where wave-in-deck loads are expected to be a problem, should include simulation of dynamic structural response under the influence of extreme waves. Wave-in-deck loading may bring the response into the nonlinear domain [1].

For a dynamic analysis of fixed offshore jacket structures exposed to wave-in-deck loading it is not evident which wave (-history) to use. In a static analysis, a worst-case scenario is used, e.g. a 100 years wave (ULS situation, note that a 100 years wave is the wave with an annual probability of exceedance of 10^{-2}) or a 10 000 years wave (ALS situation) with corresponding periods. Typical wave heights and periods for return periods of 100 and 10 000 years in the southern and northern North Sea, respectively, can be as follows [2,3]:

Southern North Sea		Northern North Sea	
$h_{100} = 26\,\text{m}$	$T_{100} = 15\,\text{s}$	$h_{100} = 28\,\text{m}$	$T_{100} = 15.5\,\text{s}$
$h_{10\,000} = 33\,\text{m}$	$T_{10\,000} = 16\,\text{s}$	$h_{10\,000} = 35\,\text{m}$	$T_{10\,000} = 16.3\,\text{s}$.

In the static analysis, these design waves 'cover' all smaller waves. In a dynamic analysis, however, a smaller wave with a period that could cause dynamic amplification could theoretically be more onerous, resulting in higher load effects. For an impact load, the form and duration of the load impulse are of main importance [4]. The load history prior to the extreme wave also influences the dynamic response.

1.1. Wave-in-deck load models

There is no general consensus on which method to use to calculate wave loads on platform decks. Several approaches exist, some verified against experimental data, some not. The methods can be divided into two main groups; the global or the silhouette approaches, which use an effective deck area exposed to the pressure from the water particles, and the detailed component approaches where loads on single members are calculated separately. A brief overview of the methods are given in the following. For details, reference is made to [1].

In the *component approaches* one seeks to estimate wave loading on each deck member and all equipment separately, requiring a computer program to model the deck in detail and to carry out the calculations.

- *Kaplan's* [5] model uses stretched [6] linear wave theory.
- *Finnigan and Petrauskas*' model [7], denoted 'Chevron model' in the comparative study[1] by HSE [8], is based on regular Stream function wave theory and Morison equation. Only horizontal loads are addressed.
- *Pawsey et al.* [9] developed a procedure based on Kaplan's recommendations but modified it to use Stream function wave theory. The integration of the wave-in-deck load module into the wave-load generator in the analysis program is emphasised.

[1]The British Health and Safety Executive has conducted a comparative study of wave-in-deck load models [8], comprising the API model, DNV slamming model, Kaplan model and in-house oil company models.

The *silhouette models* are based on an equivalent deck area and assumptions regarding the water pressure on that area. They can be divided into two different formulations, of which the first is the *drag formulation* known from e.g. Morison equation, estimating the loading as

$$F_x = \frac{1}{2}\rho A C_d u_w^2, \tag{1}$$

where ρ is seawater density, A is the exposed area and u_w is the water particle velocity. The drag factor C_d is chosen to account for different loading scenarios. Note that time variation of the loading has not been addressed for any of the models listed in the following, and accordingly they have not been subject to time domain comparison using e.g. more detailed methods or experiments. The drag formulation comprises the following models, of which the main differences are the choice of drag factor and the definition of u_w:

- *The API model* [10,11,7].
- *The ISO procedure* [12] (directly adopted from API, see above).
- *Det Norske Veritas (DNV)*[2] [16].

The second formulation categorised as a *silhouette approach* is the *momentum formulation*. This type of formulation is based on the assumption of complete loss of momentum at impact, which in a general manner can be expressed as

$$F_x(t) = \int_{A(t)} \frac{dm}{dt} u_w(z,t)\, dA, \tag{2}$$

where dm/dt is the net mass flow imparted onto the structure pr. unit time and unit area and u_w is the water particle velocity, A is the exposed area, which is a function of the surface elevation η, which again is a function of time. The formulation makes it relatively convenient to calculate load as a function of time.

- *Shell formulation*: The only available information about this method is that given by the comparative study conducted by HSE [8]. The net mass flow to be substituted into Eq. (2) for wave impact on the front wall is expressed as

$$\frac{dm}{dt} = \rho u_w(\eta), \tag{3}$$

  where ρ is the sea water density.
- *The MSL* method [17,18] is developed from the Shell method, and is intended for hull-like decks.
- *Vinje* [19] suggests that the net mass flow is dependent upon the wave celerity $c = L/T$, as opposed to the water particle velocity:

$$\frac{dm}{dt} = \rho c \tag{4}$$

[2]This formulation is used by [13,14], and is referred to as the 'Statoil method' in the comparative study conducted by HSE [8]. This formulation was originally intended for calculating vertical loads on horizontal cylinders (braces), for wedge entry into water and for flat bottom slamming. Clearly, the validity related to calculation of wave-in-deck loads can be questioned. This issue has been addressed by Vinje [15]. The conclusion is that the identification of a proper slamming coefficient is a problem, and that this drag type formulation is not suitable for calculation of wave-in-deck loading.

The main differences between the Shell/MSL method and the Vinje method are the definition of net mass flow onto the deck, and the fact that the Shell method includes the generation of loads as the wave travels along the deck.

2. Comparison of drag and momentum formulations (Eqs. (1) and (2))

The ratio of a general drag formulation (subscript 'dr') to the Shell or MSL type of formulation (subscript 'mo') is *constant* throughout the wave cycle [1]:

$$\frac{F_{\mathrm{dr}}(t)}{F_{\mathrm{mo}}(t)} = \frac{C_{\mathrm{d}}}{2} \tag{5}$$

This means that *a drag formulation with an equivalent drag factor* $C_{\mathrm{d,eq}} = 2$ *will give the same result as a momentum formulation with* $\mathrm{d}m/\mathrm{d}t = \rho u_{\mathrm{w}}$. The ratio of a general drag formulation (subscript 'dr') to the Vinje type of momentum formulation (subscript 'Vi'), however, *varies* through the wave cycle [1]:

$$\frac{F_{\mathrm{dr}}(t)}{F_{\mathrm{Vi}}(t)} \leqslant \frac{C_{\mathrm{d}} u_{\mathrm{w},max}}{2c} \tag{6}$$

In the following, we will look at the variation in drag and Vinje (momentum) formulation with time, choosing a drag factor C_{d} for the drag formulation deliberately in order to calibrate the maximum load obtained by the drag formulation to the maximum load obtained by the Vinje formulation:

- Drag formulation (Eq. (1)), $C_{\mathrm{d}} = 4.02$ / Stokes fifth theory.
- Vinje momentum formulation (Eqs. (2), (4)) / Stokes fifth theory.

A wave with $h = 33\,\mathrm{m}$ and $T = 15\,\mathrm{s}$ is used, in a water depth of $d = 75\,\mathrm{m}$. According to Stokes fifth order theory the crest height is 20.98 m. The deck freeboard z_{d} is chosen to give a deck inundation of 0.5 m.

In Fig. 1 the results are shown. The Vinje approach does not differ notably from the drag formulation for the chosen C_{d}. The differences are 0–2%, see Fig. 2(a). An equivalent drag factor as a function of time can be obtained by solving $F_{\mathrm{dr}} = F_{\mathrm{Vi}}$ with respect to C_{d}. The result is shown in Fig. 2(b). $C_{\mathrm{d,eq}}$ varies from 4.016 to 4.102.

With the small differences between the drag and the two momentum approaches in mind, the question of which formulation to use—drag or momentum—becomes less

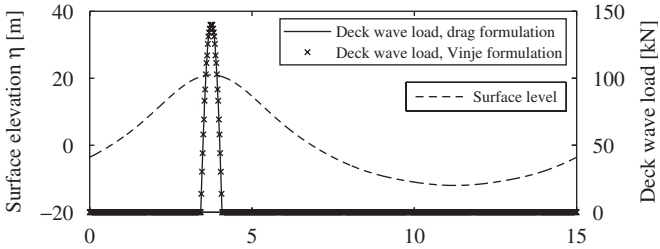


Fig. 1. Comparison of simplified wave-in-deck calculations.

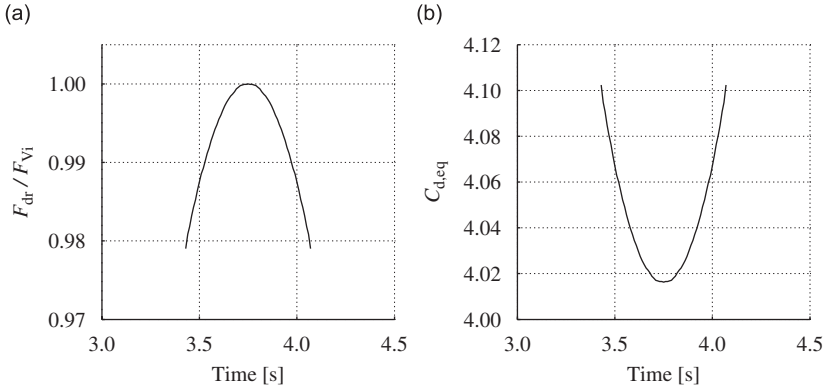


Fig. 2. Equivalent drag factors for drag formulations: (a) $C_{d,eq}$ Airy theory; (b) $C_{d,eq}$ Stokes fifth order theory.

important. Instead, the relevant question becomes which drag factor to use, alternatively which definition of rate of mass to use, $dm/dt = \rho u_w$ or $dm/dt = \rho c$.

3. Comparison of the simplified methods with computational results reported by Iwanowski et al. [20]

This section documents the comparison of load time histories obtained using the simplified methods described in the previous sections with load time histories computed with more advanced methods. Note that the methods discussed previously include loads on the front wall only.

Based on the conclusion in the last paragraph in the previous section, it is chosen to calculate the wave-in-deck load time history in two different ways:

- Traditional momentum approach with $dm/dt = \rho u_w$ (this is identical to drag formulation with $C_d = 2$), in the following denoted 'Mom', and with use of Stokes fifth order wave theory.
- Momentum formulation with $dm/dt = \rho c$ (Vinje approach), in the following denoted 'Mom-Vinje', and with use of Stokes fifth order wave theory.

Iwanowski et al. [20] presented and compared wave-in-deck load time histories calculated by use of three different software. Only the calculations applying Stokes wave are used for comparison purposes herein, meaning use of CFD programs FSWL-2D and FLOW-3D, the latter in both 2D and 3D modes.

The calculations were carried out for a 100 years design wave for the Ekofisk field in the Southern North Sea with the characteristics $h = 24.3$ m, $T = 14.5$ s and $d = 80$ m.

3.1. Deck modelled as a simple box, 2 m inundation

The wave loads were calculated for a deck modelled as a simple box being 30 m wide (normal to the wave propagation direction), with wave inundation 2 m [20].

The Iwanowski load histories are in Fig. 3 compared to the load histories calculated in the present project.

The loads calculated by Iwanowski et al. for the Stokes wave all show quite similar trends. The start and end time for the loads are essentially the same, and the maximum values range from 2.8 to 3.5 MN. The Vinje formulation yields a maximum load of 10.14 MN. This is considerably larger than the values computed by Iwanowski et al. The explanation could be that Vinje's formula assumes that the horizontal water momentum is being stopped by the deck while some water particles in practice are being distorted upon impact with the deck. The drag formulation with $C_d = 2$, however, has a maximum load of 3.4 MN which agrees well with the Iwanowski results. The shapes of the impulses are similar, however the CFD-results are somewhat skewed towards the start time, while the simplified approaches by their nature produce symmetric load histories.

3.2. Simplified deck geometry, Ekofisk 2/4 C platform

Iwanowski et al. also calculated the wave loads for a simplified deck geometry consisting of a lower box measuring 42.6 m × 30 m × 1.5 m centrally attached to an upper box measuring 53.1 m × 42 m × 10 m (all measures given as length × width × height, where width is measured normal to the wave heading). The wave inundation is 1.5 m, i.e. reaching but not entering the 'floor' of the upper box. A deck width of 30 m is therefore used for calculation of loads by the simplified methods. A comparison is illustrated in Fig. 4.

For this case, the CFD methods compute a relatively large peak load, which agrees well with the Vinje formulation. Water gets trapped in the corner between the lower and the upper box, and the CFD techniques are able to simulate the local fluid flow in this corner with the high peak as a result. The simplified drag- and momentum approaches are, as already mentioned, only capable of predicting symmetric load histories on the front wall.

The sharp load peak that characterises the CFD results has a duration of about half a second, in which the load rapidly increases from zero to maximum and decreases to about $\frac{1}{5}$ of the maximum load. Thereafter, the load further decreases more slowly within about half a second to zero, or temporarily somewhat below zero (i.e. suction load).

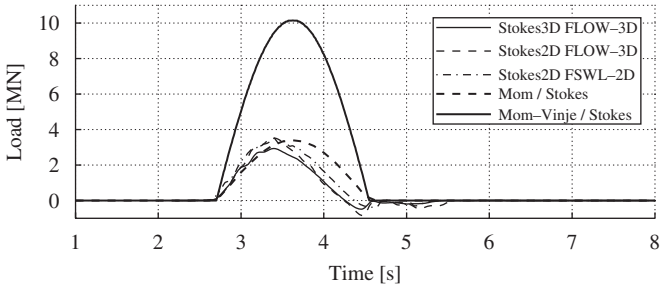


Fig. 3. Comparison of simplified calculations and Iwanowski results for simple box, inundation 2 m, deck width 30 m.

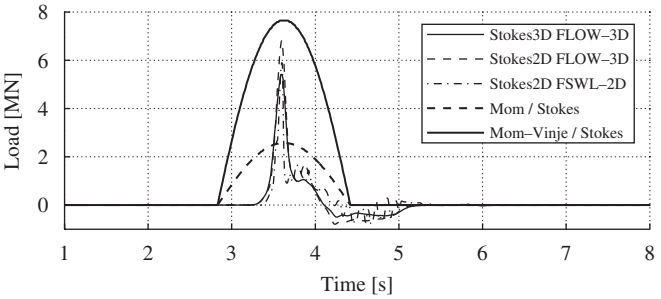


Fig. 4. Comparison of simplified calculations and Iwanowski results for a simplified Ekofisk 2/4 C deck geometry, inundation 1.5 m.

3.3. Summary of Section 3

Clearly, the simplified methods presented herein are not able to accurately predict wave-in-deck load histories for detailed platform decks. However, a representative load history for a simple hull type of deck can probably be produced. It is, however, always important to consider the objective of the analyses. For detailed (re-)analyses that are meant to document the performance of specific structures, simplified methods may not be adequate.

4. Published wave-in-deck experimental results

Wave-in-deck loads are sensitive to wave inundation, i.e. crest height. During experiments with regular waves, the crest heights of measured waves may vary much more than the wave heights [21]. When comparing measured and numerically predicted data, it is therefore important to pay attention to crest heights, not only wave heights.

'Gulf of Mexico related' experiments reported in [7] have been used to calibrate the API procedure and the Chevron procedure. Deck loads at only one time instant per experiment, probably at maximum load, are reported. This coincides well with the fact that the API and the Chevron procedures aim at estimating the maximum loads for static structural analysis, but unfortunately this makes the results unsuited for time history considerations.

Results from experiments carried out at the large wave channel (*der Grosse Wellenkanal*) at Forschungszentrum Küste in Hannover are published in [21]. The experiments comprise wave load time histories on typical offshore deck elements, both single elements and element groups, and focus has been on the details of the loading process. However, since no results are published for complete deck models the results are unsuitable for comparison with assumptions made in the present work.

Early in 2002, model tests at scale 1:54 were carried out at Marintek in Trondheim in connection with a possible late life production scenario for the three shaft concrete gravity base structure (GBS) platform *Statfjord A* [22]. Global deck loads and local slamming loads were, amongst others, measured. The results from the model tests have been interpreted in a report from Marine Technology Consulting AS to Statoil [23], and recommendations are given regarding which time histories to use for wave-in-deck

slamming load when carrying out structural analyses of Statfjord A in case of seabed subsidence possibly caused by reduced reservoir pressure.

5. Results from wave tank experiments—Statfjord A GBS

The experiments were carried out for two water depths: 150.1 and 151.6 m. This corresponds to 0.5 and 2.0 m inundation for the 10 000 years crest of 21.7 m. The impact loads relating to this crest in these two water depths were estimated to be 75 and 105 MN, respectively, and these values were recommended for design checks. In order to determine a representative load time history, a selection of measured time histories was investigated. For 150.1 m water depth only measured load time histories with maximum load between 50 and 100 MN were considered, in total 31 time histories. In the same manner, only time histories having maximum loads ranging from 80 to 125 MN were investigated for a water depth of 151.6 m, this left 22 time histories.

In Table 1, the recommended horizontal maximum loads $F_{d,max}$ and the load at the kink F_k (see definition in Fig. 12) from the Statfjord experiments are shown, together with the computed results from [20], note that only the computer results obtained by Stokes wave and FLOW-3D program in 3D mode are shown.

There are, besides the fact that neither the inundation level nor the deck width are the same, several reasons that the numbers in Table 1 cannot be directly compared. These issues are discussed in the following:

(1) Statfjord A is a GBS platform with a huge base supporting three large diameter columns. Both the presence of the base as well as the reflection of waves from the columns and the interaction between the reflected and next incoming wave result in amplification of the incoming wave. Stansberg et al. [22] indicate that approximately a 20% amplification of the wave height compared to a (undisturbed) regular 30 m wave with periods of some 16.5 s. Wave-in-deck loads are reported as a function of the crest height for the *undisturbed wave*, however they are actually generated by an *amplified wave*. As a consequence of this, the real inundation is greater than the value reported in [23] or in the above Table 1. In fact, some waves that in undisturbed condition do not enter the deck do also, when amplified, generate wave-in-deck loads.
This may explain the small increase in load, and the corresponding reduced water

Table 1
Reported horizontal wave-in-deck loads

Reference	Iwanowski	Statoil	Statoil
Type of results	CFD 2/4 C deck	Experiments	Experiments
Wave	100 years	10 000 years	10 000 years
Inundation	1.5 m	0.5 m[a]	2.0 m[a]
Deck width	30 m	83.6 m	83.6 m
$F_{d,max}$	5.4 MN	75 MN	105 MN
Pressure due to $F_{d,max}$[b]	0.12 MN/m^2	1.79 MN/m^2	0.63 MN/m^2
F_k	N/A	30 MN	35 MN

[a]Note that the inundation is calculated from *undisturbed* wave crest height.
[b]On the inundated area.

pressure on the inundated area, for the 2.0 m inundation case in the Statoil experiments compared to the 0.5 inundation case. Now considering the increase in load caused by the increased inundation; the 1.5 m increase in inundation corresponds to the load being increased by 30 MN. The pressure cause by this increase is $30/(1.5 \cdot 83.6)\,\mathrm{MN/m^2} = 0.24\,\mathrm{MN/m^2}$. This measure might be a better indication of the water pressure caused by a wave that is not subject to amplification by the presence of the substructure, which is the case for waves acting on jacket platforms.
Ekofisk 2/4 C (which is the structure investigated by Iwanowski et al.) is a jacket platform, for which the wave amplification due to the presence of the structure itself is negligible. As a consequence, the load generated by the increased wave crest for Statfjord A cannot without discussion be compared to the load calculated for the deck of the jacket platform.

(2) The Iwanowski results are obtained for a *100 years wave*, whereas the Statoil experiments were carried out in order to find the load time history for a *10 000 years wave*. The former has smaller particle velocity in the crest, and this is obviously reflected in the calculated loads. NORSOK [3] recommends the 10 000 year design wave height to be 25% larger than the 100 year wave height. This leads to, for Southern North Sea conditions (see Section 1), an increase in the crest particle velocity of some 35%. Assuming that the particle velocity enters squared into the load, a 10 000 years Ekofisk wave is estimated to give a pressure on the inundated area of $0.22\,\mathrm{MN/m^2}$. This value corresponds well with the pressure calculated under item 1 above.

(3) The definition of *crest front steepness* used during interpretation of the Statfjord experiments is $s = \eta_{max}/(c \cdot (0.25T)) = 4\eta_{max}/L$. For the 100 years wave used by Iwanowski this steepness formulation gives $s = 0.18$. From the waves generated during the Statfjord experiments, about $\frac{3}{4}$ have crest front steepness larger than 0.3. Thus the majority of the waves forming the background for the estimate of wave-in-deck load for Statfjord A are considerably steeper than the wave used by Iwanowski.

The general trend for the global deck load is that the normalised time history for the horizontal slamming load consists of three linear parts as shown in Fig. 5. It is characterised by a steep linear rise to maximum load $F_{\mathrm{d},max}$, a steep linear decrease to about $F_{\mathrm{k}} = 0.4$ times the maximum value, and finally a less steep but still linear decrease to zero. The durations for the three phases are 0.54 s, approximately 0.5 s, and 2.1 s, respectively.

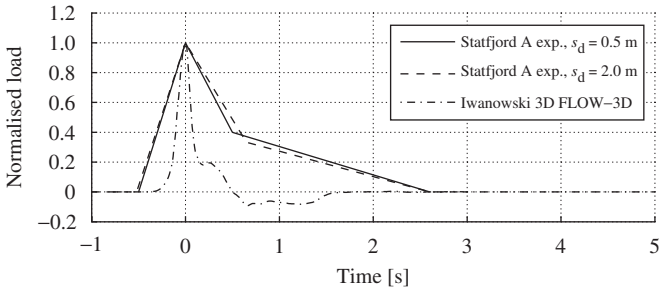


Fig. 5. Horizontal wave-in-deck load history, trend from experiments [23] and computational results [20].

It should be noted that this load time history represents a number of experiments in which the numerical values differ considerably. However, the three-line-trend is seen in most of the experiments. A single experimental wave-in-deck load time history reported by [24] supports this finding. The time history is recorded at the deck, which consists of cylindrical elements, beam and plate elements and a solid top and bottom plate, during model tests of Ekofisk 2/4 C. The three-line-trend is also seen in [20] where CFD technique is used to calculate wave-in-deck loads on a simplified platform deck.

6. Sensitivity study—deck load duration

We have carried out a study to investigate the sensitivity of the structural response to the duration of the load that acts on the platform deck. Time domain analyses of a jacket model denoted 'DS' subjected to extreme wave loading have been carried out using the finite element program USFOS. The analyses are based on previously published analyses of the same model, and for further details than given herein, reference is made to [1].

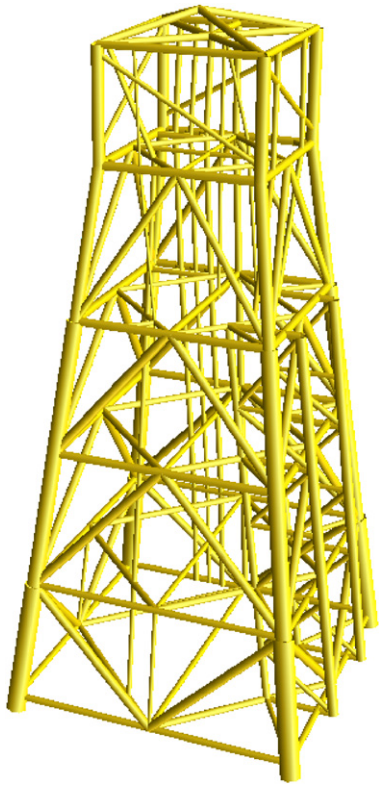


Fig. 6. Structural model 'DS'.

6.1. Structural model of the jacket 'DS'

6.1.1. General

The model jacket 'DS' (Fig. 6) is based on an existing static linear analysis model of an existing North Sea jacket, which is pile supported, K-braced and has five risers and four caissons. The area between the deck legs is 22 m × 22 m. The water depth at the field is 70 m.

The model has been somewhat simplified; for simplicity, the platform legs have been fixed to the seabed for all six degrees of freedom, and the deck structure has been replaced by a simple but stiff dummy deck structure. The deck is assumed to be 47 m × 47 m.

In the analysis model, the lowest deck is located at $z = z_d = 95.5$ m. The model coordinate system is right-handed and has its origin at the seabed. In order to simulate subsidence of seabed and the structure, the z-value of the sea surface is set differently from one analysis (load) scenario to another.

The model structure has a first natural period T_n of 1.60 s.

6.1.2. Materials and cross sections

Two different materials have been used, one typical steel material and one dummy material with higher stiffness but very small density. The latter is used for the deck dummy structure, and the former for the rest of the structure. The yield stress is 355 N/mm^2.

The diameter of the circular members range from 0.457 to 3 m and wall thickness from 0.020 to 0.095 m.

6.2. Loading

Self weight: The self weight of all members is generated automatically and sums up to 3.78×10^6 kg. In addition, a node mass of 11×10^6 kg representing the deck weight and weight of equipment and personnel is applied at a node located at the centre of gravity of the deck structure.

Wind: No wind loads are included in the analyses in order to highlight the effects of the wave loads.

Wave loads on the jacket structure: The wave load histories are generated by the USFOS program. Stoke fifth order theory [25] is used, and the structure is subjected to one wave cycle. The load histories are based on a wave with and annual probability of exceedance of 10^{-4} (a 10 000 years wave), with height $h = 33$ m and period $T = 16$ s. Since subsidence is the main trigger of wave-in-deck loading in the North Sea, two water depths have been analysed; $d = 77$ m and $d = 79$ m. Tide and storm surge is assumed to be included in the different water depths.

Wave load on the deck structure: The wave-in-deck loading is applied in accordance with the herein recommended wave-in-deck load time history as given in Fig. 12 (note that in accordance with the intentions of these analyses, the total duration of the deck load is varied). The deck load is applied to the top of the deck legs and distributed equally, meaning $\frac{1}{4}$ to each leg. Vertical effects due to wave-in-deck loading or due to buoyancy when the wave submerges the deck are not taken into account in the analyses herein.

The peak horizontal wave in deck load is assumed to occur when the wave crest is at the deck front wall at $t = 4.1$ s. As *a basis load duration*, in the following denoted $t_{d,basis}$, we have used the duration for the wave crest from when it first contacts the deck front wall to

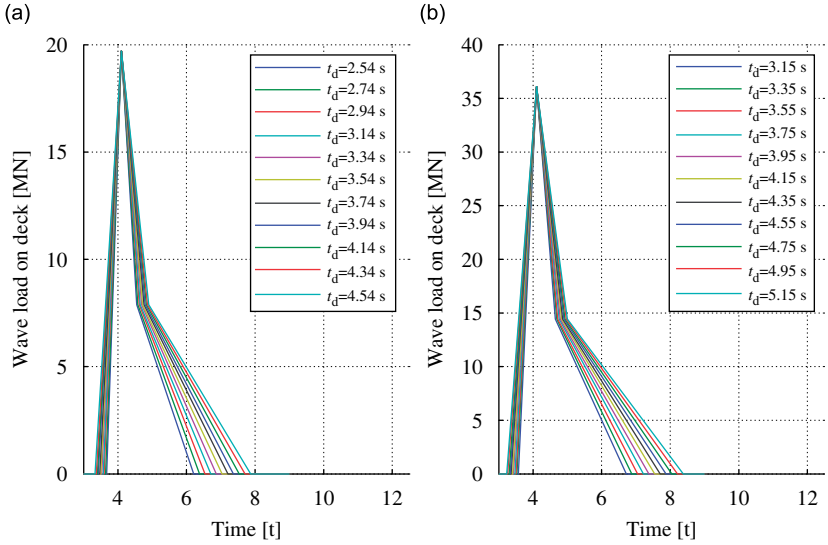


Fig. 7. Wave load time histories for the deck: (a) $h = 77$ m; (b) $h = 79$ m.

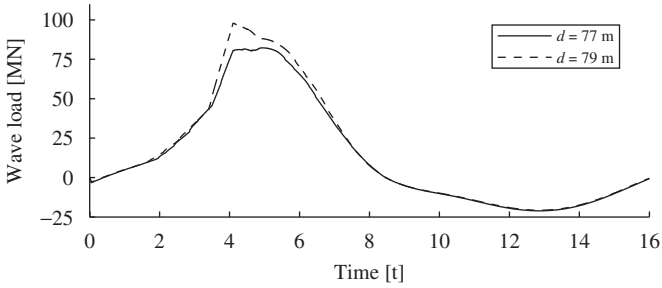


Fig. 8. Hydrodynamic load histories.

when it has travelled through the deck and finally lost contact with the deck on the opposite side, i.e. the total time of contact between the wave and the deck:

- For $d = 77$ m: $t_{d,basis} = 3.54$ s.
- For $d = 79$ m: $t_{d,basis} = 4.15$ s.

Analyses are carried out for load durations $t_{d,basis} \pm 1$ s, in steps of 0.2 s. The load time histories for the deck are illustrated in Fig. 7. These load time histories are added to the time histories of wave induced load on the jacket structure, see Fig. 8. The reference load values $F_{d,max}$ for the wave-in-deck loading are given in Table 2.

Note that for $d = 77$ m the maximum total load will occur at approximately $t = 5$ s—i.e. not simultaneously with the peak wave-in-deck load at $t = 4.1$ s—due to the relatively

Table 2
Model 'DS'; wave-in-deck loads to be used in analysis

Water depth d (m)	Crest η_{max} (m)	Deck inund. s_d (m)	$F_{d,max}$ (MN) (Fig. 12)	F_k (MN) (Fig. (12))
77.0	20.62	2.12	19.71	7.884
79.0	20.50	4.00	36.09	14.43

Table 3
Horizontal displacement response at deck level and deck load impulse, water depth $d = 77$ m

t_d (s)	u_m (m)	$u_m/u_{m,basis}$	$u_m/max(u_m)$ (s)	I (MNs)	I/I_{basis}
2.54	0.310	0.979	0.979	16.99	−0.280
2.74	0.313	0.987	0.987	18.25	−0.227
2.94	0.315	0.995	0.995	19.67	−0.167
3.14	0.316	0.998	0.998	20.93	−0.114
3.34[a]	0.317	1.000	1.000	22.35	−0.053
3.54[a],[b]	0.317	1.000	1.000	23.61	0.000
3.74	0.317	0.999	0.999	24.97	0.058
3.94	0.315	0.995	0.995	26.29	0.114
4.14	0.314	0.991	0.991	27.65	0.171
4.34	0.313	0.987	0.987	28.91	0.225
4.54	0.310	0.979	0.979	30.33	0.285

[a]Basis load duration $t_{d,basis}$.
[b]Load duration corresponding to max response.

small magnitude of the wave-in-deck load as compared to the load on the jacket structure (see also Fig. 8).

The hydrodynamic load histories including wave-in-deck loading (referred to the basis load duration $t_{d,basis}$) and current loading are shown in Fig. 8 for the analysed water depths. The wave crest is at the deck front wall at $t = 4.1$ s. The load peaks at this time instant represent the wave-in-deck loading.

Current: The following current profile is used in the analyses:

z (m)	Velocity scaling factor
0.0	1.00
−25.0	0.52
−85.0	0.28

Between these specified values of the velocity, linear interpolation is used. Above still water level ($z = 0$ m) the values are extrapolated, resulting in a varying surface current through the wave period.

Buoyancy: The jacket legs, pile sleeves, risers and caissons are flooded. Buoyancy is calculated for non-flooded elements if submerged. The buoyancy loads are included in the self weight load case, meaning it is applied as a permanent, static load. Buoyancy of the deck when impacted is not accounted for.

Table 4
Horizontal displacement response at deck level and deck load impulse, water depth $d = 79$ m

t_d (s)	u_m (m)	$u_m/u_{m,basis}$	$u_m/max(u_m)$ (s)	I (MNs)	I/I_{basis}
3.15	0.555	0.934	0.931	38.40	−0.243
3.35	0.566	0.953	0.950	41.00	−0.191
3.55	0.574	0.967	0.964	43.31	−0.146
3.75	0.584	0.983	0.980	45.80	−0.097
3.95	0.588	0.989	0.986	48.22	−0.049
4.15[a]	0.594	1.000	0.997	50.71	0.000
4.35	0.595	1.001	0.998	53.12	0.048
4.55[b]	0.596	1.003	1.000	55.61	0.097
4.75[b]	0.596	1.003	1.000	57.92	0.142
4.95	0.593	0.999	0.995	60.52	0.194
5.15	0.589	0.992	0.989	62.83	0.239

[a]Basis load duration $t_{d,basis}$.
[b]Load duration corresponding to max response.

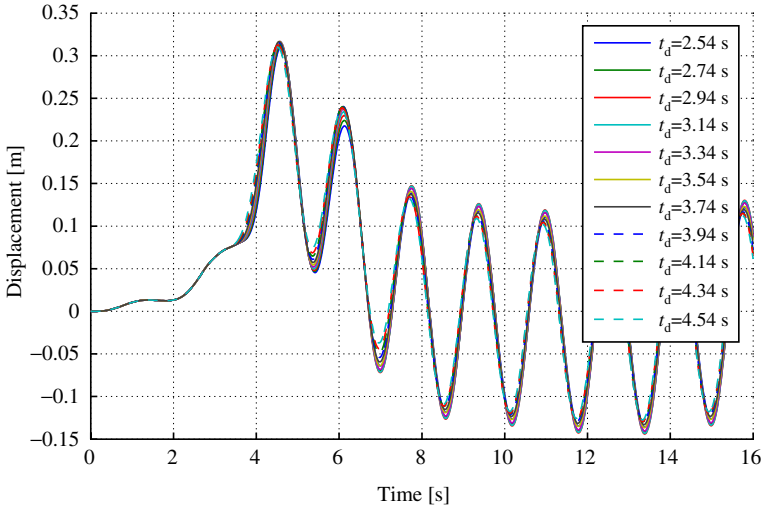


Fig. 9. Horizontal displacement response at deck level, water depth $d = 77$ m.

6.3. Results from the analyses

The results from the sensitivity study is summarised in Tables 3 and 4 and Figs. 9 and 10. It is clear from these that the response is not very sensitive to the duration of the part of the wave load that acts on the deck *within the ranges analysed herein*.

For the $d = 77$ m case (Table 3 and Fig. 9), the largest response u_m (horizontal displacement measured at deck level) ranges from 0.310 to 0.317 m.

The latter is 2.3% larger than the former. The basis load duration ($t_{d,basis} = 3.54$ s) is obviously a good choice for the load duration in this case.

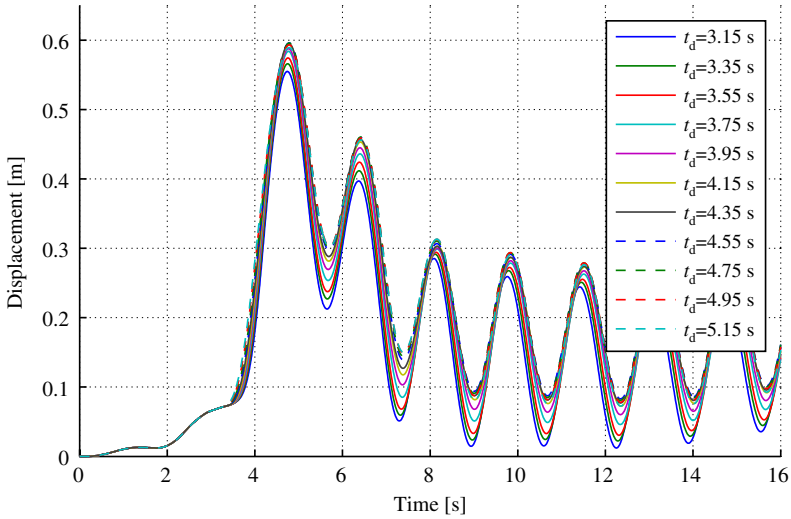


Fig. 10. Horizontal displacement response at deck level, water depth $d = 79$ m.

The deck load impulses (denoted I) vary from 17.0 to 30.3 MNs. This is some ±29% compared to the impulse relevant for the basis load duration ($t_{d,basis} = 3.54$ s), for which $I = 23.6$ MNs.

For the $d = 79$ m case (Table 4 and Fig. 10), the maximum displacement response u_m ranges from 0.555 to 0.596 m. For this water depth, the basis load duration is $t_{d,basis} = 4.15$ s, while the largest response occurs for $t_d = 4.55$ s and $t_d = 4.75$. The largest response is 0.3% larger than the response related to $t_{d,basis}$, which also for this water depth is a representative choice for the load duration.

The deck load impulses (I) vary from 38.4 to 62.8 MNs. This corresponds to ±24% compared to the impulse relevant for the basis load duration ($t_{d,basis} = 4.15$ s), for which $I = 50.7$ MNs. For the load duration that corresponds to the maximum response ($t_{d,basis} = 4.55$ s and $t_d = 4.75$), the impulse is 55.6 MNs, an increase of some 10% compared to $t_{d,basis}$. This, however, yields only a slight increase in the response peak, as mentioned.

For the two water depths analysed herein, the basis load duration[3] are good estimates of the duration of the wave-in-deck load history.

7. Vertical loads

Till now, only horizontal loads have been considered. However, a few of the referred publications have treated vertical load time histories as well [20,23].

In the Statfjord A wave tank experiment [23], wave loads were measured and interpreted. Recommendations for wave-in-deck loading in the form of reference loads (max and min) and time history shape for reassessment of Statfjord A GBS were given.

[3]Defined as the duration for the wave crest from when it first contacts the deck front wall to when it has travelled through the deck and finally lost contact with the deck on the opposite side, i.e. the total time of contact between the wave and the deck.

Table 5
Reported vertical wave-in-deck loads

Reference	Iwanowski	Statoil	Statoil
Type of results	CFD 2/4 C deck	Experiments	Experiments
Wave	100 years	10 000 years	10 000 years
Inundation	1.5 m	0.5 m[a]	2.0 m[a]
Deck width	30 m	83.6 m	83.6 m
$F_{v,max}$	41 MN	67 MN	80 MN
$F_{v,min}$	−22 MN	−35 MN	−50 MN

[a]Note that the inundation is calculated from *undisturbed* wave crest height.

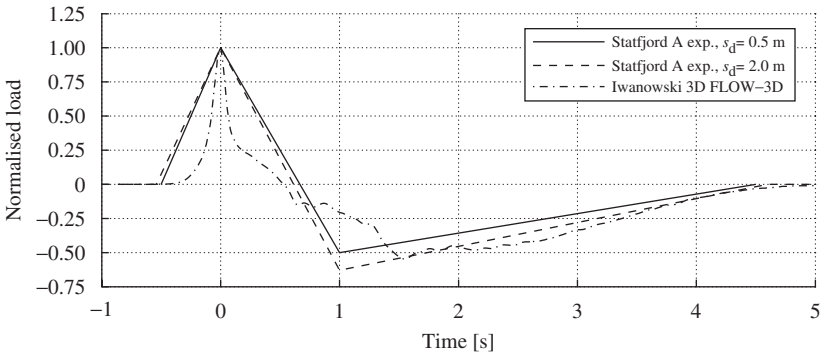


Fig. 11. Vertical wave-in-deck load history, trend from experiments [23] and computational results [20].

The horizontal loads in the Statoil report range from 50 to 100 MN for $d = 150.1$ m and from 80 to 125 MN for $d = 151.6$ m, with recommended design values 75 and 105 MN, respectively. The recommended maximum positive vertical loads are somewhat smaller than the horizontal loads, 67 and 80 MN, respectively. The minimum load, which is negative (suction), is about 50–60% the value of the maximum vertical load (the deck width is 83.6 m). There is however, considerable uncertainty connected to these numbers, and they should only be regarded as rough but indeed representative outline of the observed vertical wave-in-deck loading. The load values that were recommended for design are shown in Table 5 together with the loads from the CFD results reported by Iwanowski et al. for 30 m deck width. Note that only the CFD results obtained by Stokes 3D FLOW-3D are used.

The recommended (normalised) load time history from the Statoil report is characterised by a linear rise from zero to maximum, with duration of about 0.5 s, thereafter a linear drop to minimum load, which is negative, in about 1 s. Finally, the load increases linearly from its minimum to zero in about 3.5 s. This recommendation is given on background of 31 measured load histories, to which a representative load time history was fitted by means of least square method. The Statoil recommendation is compared to the Iwanowski CFD results for the simplified 2/4 C deck in Fig. 11. All time histories are normalised against their respective maximum loads. The time variation of the vertical load is in practice the same for these two independent studies, of which one is theoretical and the other one experimental.

It is concluded that vertical wave-in-deck loads are of considerable magnitude, and act both upwards and downwards. They result in deck uplift loads, and they give additional compressive loads in platform legs, which can lead to different failure modes than the platform originally was designed to sustain. *It is emphasised that vertical loads should be considered during reassessment of offshore platforms, however, this topic is not further treated herein.*

8. Preceding load time history

This issue is not treated in this article, however a number of authors have given recommendations for how long pre-history should be included in the wave time history before the extreme wave, see [26], section on 'representative load histories'.

9. Discussion

9.1. Time history

The three-line (load) time history referred to above is the type of deck load history for which most support is found. We recommend to use this type of load history, see Fig. 12. The load time history is described in full by this time history and a reference load, taken as the maximum load $F_{d,max}$ corresponding to the current inundation level.

The values on the horizontal axis are given relative to the basis load duration $t_{d,basis}$, defined as the duration for the wave crest from when it first contacts the deck front wall to when it has travelled through the deck and finally lost contact with the deck on the opposite side, i.e. the total time of contact between the wave and the deck.

The sensitivity to the deck load duration is investigated for two different water depths and corresponding inundation levels. The basis load duration $t_{d,basis}$ is found to be a representative choice for the deck load duration, as it yields the largest or close to the largest response peak (displacement) in the study. For deck load durations close to $t_{d,basis}$, the change in response peak u_m is negligible.

It should be noted, though, that this conclusion may not be valid for larger inundation levels.

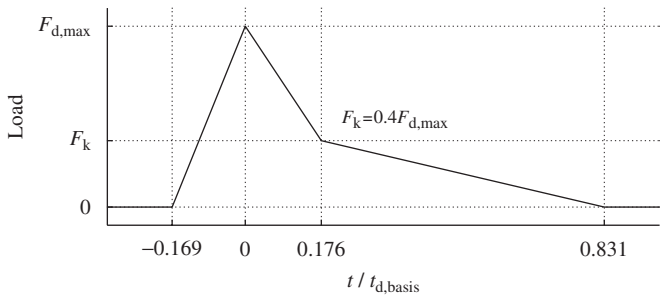


Fig. 12. Recommended load time history for use in analyses.

Table 6
Wave-in-deck loads, $s_d = 1.5$ m and $b = 30$ m

Reference	Wave type	H (m)	T (s)	d (m)	c (m/s)	u_{cs} (m/s)	u_{ce} (m/s)	$F_{d,max}$ (MN)	$p_{d,max}$ (MN/m^2)
Statfjord A	10 000 year, measured (basis)	36.5	15.8	150		9.80	0	5.87	0.13
Statfjord A	100 year, Eq. (8)	29.0	14.4	150		8.25	0	4.16	0.09
Statfjord A	10 000 year, API[b]	36.5	15.8	150		9.80	0	4.43	0.10
Statfjord A	10 000 year, Vinje formulation	36.5	15.8	150	26.17	9.80	0	11.83	0.26
Statfjord A	10 000 year, + curr., Eq. (8)	36.5	15.8	150		9.80	1	7.13	0.16
Statfjord A	10 000 year, + curr., API[a]	36.5	15.8	150		9.80	1	5.38	0.12
Iwanowski	100 year, calc. Iwanowski[b]	24.3	14.5	80		N/A	N/A	5.40	0.12
Iwanowski	100 year, Eq. (8)	24.3	14.5	80		7.57	0	3.50	0.08
Iwanowski	100 year, API[a]	24.3	14.5	80		7.57	0	2.64	0.06
Iwanowski	100 year, Vinje formulation	24.3	14.5	80	22.22	7.57	0	7.76	0.17
SNS[c]	100 year, Eq. (8)	26.0	15.5	75		8.17	0	4.08	0.09
SNS	100 year,API[a]	26.0	15.5	75		8.17	0	3.08	0.07
SNS	10 000 year, Eq. (8)	33.0	16.0	75		11.28	0	7.77	0.17
SNS	10 000 year, API[a]	33.0	16.0	75		11.28	0	5.87	0.13
SNS	10 000 year, Vinje formulation	33.0	16.0	75	23.75	11.28	0	12.36	0.27
SNS	10 000 year + curr., Eq. (8)	33.0	16.0	75		11.28	1	9.21	0.20
SNS	10 000 year + curr., API[a]	33.0	16.0	75		11.28	1	6.96	0.15

[a]Using $C_d = 2.0$, corresponding to moderately equipped deck, end on/broad side loading.
[b]Calculated by use of CFD methods [20].
[c]Southern North Sea platform.

9.2. Reference load $F_{d,max}$

The load level for an actual wave-in-deck situation depends on the local geometry of the deck. We will, however, recommend using the regression curves obtained from the experimental data during the Statfjord A experiments as a basis. The experimental data for $d = 151.6$ m is split into three different *crest front steepness* ranges, and the linear regression curve for steepness $s < 0.3$ is used [22, Fig. 9], since the Stokes fifth order waves relevant for the present study will belong to this range (note that crest front steepness is defined as $s = \eta_{max}/(c \cdot 0.25T) = 4\eta_{max}/L$). From the uppermost subfigure of Fig. 9 in the given reference, the variation of wave-in-deck load with inundation is found to be 10.9 MN/m. Dividing by the deck width of 83.6 m this leaves[4] 0.13 MN/m^2. In order to omit the influence of the wave amplification over the gravity base, we suggest setting the horizontal load equal to zero for a wave crest that just reaches the underside of the deck. Larger wave crests generate loads that are proportional to the inundation with a factor of 0.13 MN/m inundation for unit deck width

$$F_{d,max} = 0.13 s_d b \,(\mathrm{MN}), \tag{7}$$

where s_d is inundation and b the deck width. This equation is related to a 10 000 years wave at the Statfjord field in the Northern North Sea, with a corresponding crest particle

[4]Note that in the argumentation following Table 1 in Section 5, the results from *all* steepness ranges are included, and therefore a higher pressure is obtained.

velocity. It is assumed that the particle velocity enters squared into the equation for the load. This is true for both a drag formulation and a general momentum formulation (but not for Vinje formulation). In order to allow for adjustment of the load to represent the actual wave and to include current, we suggest that Eq. (7) be modified as follows:

$$F_{d,max} = 0.13 s_d b \frac{(u_{cs} + u_{ce})^2}{u_{ref}^2} \text{ (MN)}, \tag{8}$$

where u_{cs} is the water particle velocity in the wave crest, u_{ce} is the current velocity and u_{ref} is the particle velocity representing the 10 000 years wave at the Statfjord field, which by use of Stokes fifth order theory is found to be 9.8 m/s. In Table 6, the reference (i.e. maximum) values for the deck load calculated by this method for several different scenarios are listed. Also included is the maximum load calculated by Iwanowski et al. [20] for the simplified Ekofisk 2/4 C deck, as well as load values calculated according to Vinje formulation and the drag formulation recommended in the API regulations with $C_d = 2.0$ (note that API recommends a drag factor between 1.2 and 2.5, where 2.0 corresponds to end-on or broadside loading of moderately equipped deck). The loads are calculated for a deck width $b = 30$ m and an inundation $s_d = 1.5$ m.

Table 6 illustrates that the API recommendations with $C_d = 2.0$ in general yields lower loads than Eq. (8) gives ($p_{d,max}$ is the average pressure on the inundated area caused by $F_{d,max}$). The fraction is about 75%. If increasing the C_d to 2.5 (end-on or broadside loading of heavily equipped/solid deck), the fraction would be $75\% \cdot 2.5/2.0 = 94\%$, i.e. Eq. (8) would still yield conservative loads compared to the API regulations.

The Vinje formulations yields much higher loads compared to the other methods.

The maximum load calculated according to Eq. (8) for the Iwanowski wave is 3.50 MN. This is considerably smaller than the value calculated in the reference paper [20] for this deck (5.4 MN). However, the API formulation yields even smaller loads, only about 50% of the value calculated by Iwanowski et al.

10. Recommendations

The above discussion is considered to support Eq. (8) being a reasonable estimate for wave load on deck for an example jacket structure. This equation together with the load history given in Fig. 12 is sufficient to establish wave-in-deck load histories for analyses of jacket wave-in-deck response.

Note that the given 'recipe' is intended for cases where a simplified estimate of deck load time history can be accepted, it is not intended to replace detailed analyses or wave tank experiments where detailed analyses are considered necessary to document the structural integrity of a specific jacket platform.

11. Further work

We have prepared a recommendation (Eq. (8)) for calculation of wave-in-deck loads on jacket structures after having discussed available data and information where we have put main emphasis on a data set available from wave-tank test of the wave-in-deck loading on an offshore platform. Validation of the recommendations through tank tests of wave-in-deck loading on jacket decks would be strongly recommended.

Acknowledgements

The authors will thank Sverre Haver of Statoil, Tor Vinje of Marine Technology Consulting AS and Professor Jasna Bogunović Jakobsen of University of Stavanger for fruitful discussions during preparation of this paper. The Statfjord Late Life project carried out by the Statfjord licence is acknowledged for permission to refer to the MTC report, Ref. [23].

References

[1] van Raaij K. Dynamic behaviour of jackets exposed to wave-in-deck forces. Ph.D. thesis, University of Stavanger (December 2005).
[2] Eik KJ. Personal communication, Statoil ASA (January 2005).
[3] NORSOK Standard N-003, Actions and action effects, 1st ed. ⟨http://www.standard.no/⟩ (February 1999).
[4] Biggs JM. Introduction to structural dynamics. New York: McGraw-Hill; 1964.
[5] Kaplan P, Murray JJ, Yu WC. Theoretical analysis of wave impact forces on platform deck structures. In: Proceedings of the 14th international conference on offshore mechanics and arctic engineering (OMAE) 1995, Copenhagen, Denmark; 1995.
[6] Wheeler JD. Method for calculating forces produced by irregular waves. J Petrol Technol 1970:359.
[7] Finnigan TD, Petrauskas C. Wave-in-deck forces. In: Proceedings of the 7th international offshore and polar engineering conference 1997, vol. III. Honolulu, Hawaii, USA; 1997. p. 19–24.
[8] HSE. Review of wave-in-deck load assessment procedures. Technical Report OTO 97 073 (MaTSU/8781/3420). Health & Safety Executive. United Kingdom, prepared by BOMEL and Offshore Design, 1997.
[9] Pawsey S, Driver D, Gebara J, Bole J, Westlake H. Characterization of environmental loads on subsiding offshore platforms. In: Proceedings of the 17th international conference on offshore mechanics and arctic engineering (OMAE) 1998. Lisbon, Portugal, 1998.
[10] API WSD, Recommended practice for planning, designing and constructing fixed offshore platforms—Working stress design (API RP2A-WSD)/Supplement 1, Washington, DC, USA: American Petroleum Institute; 1997.
[11] API LRFD, Recommended practice for planning, designing and constructing fixed offshore platforms—Load and resistance factor design (API RP2A-LRFD)/Supplement 1, Washington, DC, USA: American Petroleum Institute; 1997.
[12] ISO/CD 19902, Petroleum and Natural Gas Industries—Fixed Steel Offshore Structures (June 2001).
[13] Dalane JI, Haver S. Requalification of an unmanned jacket structure using reliability methods. In: Proceedings of the 27th annual offshore technology conference 1995, Houston, Texas, USA; 1995, oTC 7756.
[14] Haver S. Uncertainties in force and response estimates. In: Uncertainties in the design process, offshore structures/metocean workshop, Surrey, England, 1995, e & P Forum Report No. 3.15/229.
[15] Vinje T. Comments to the DNV rules regarding slamming pressures. Open note prepared for Statoil, 2002.
[16] Det Norske Veritas. Environmental conditions and environmental loads. Oslo, Norway, Classification note no. 30.5 March 1991.
[17] HSE, Assessment of the effect of wave-in-deck loads on a typical jack-up. Technical Report Offshore Technology Report 2001/034, Health & Safety Executive, United Kingdom, prepared by MSL Engineering Ltd, 2001.
[18] HSE, Sensitivity of jack-up reliability to wave-in-deck calculation, Technical Report Research Report 019, Health & Safety Executive, United Kingdom, prepared by MSL Engineering Ltd., 2003.
[19] Vinje T. Presentation given at wave-in-deck Seminar at Statoil 17 January 2001. Printed in compendium from the seminar, 2001.
[20] Iwanowski B, Grigorian H, Scherf I. Subsidence of the ekofisk platforms: wave in deck impact study. Various wave models and computational methods. In: Proceedings of the 21st international conference on offshore mechanics and arctic engineering (OMAE) 2002, Oslo, Norway; 2002.

[21] Sterndorff MJ. Large-scale model tests with wave loading on offshore platform deck elements. In: Proceedings of the 21st International Conference on Offshore Mechanics and Arctic Engineering (OMAE) 2002, Oslo, Norway; 2002.
[22] Stansberg CT, Baarholm R, Fokk T, Gudmestad OT, Haver S, Wave amplification and possible deck impact on gravity based structure in 10^{-4} probability extreme crest heights. In: Proceedings of the 23rd international conference on offshore mechanics and arctic engineering (OMAE) 2004, Vancouver, BC, Canada; 2004.
[23] Statoil, Statfjord A, slamming forces for design, Technical Report MTC-27-2, Statoil, confidential report prepared by Marine Technology Consulting AS, August 2002.
[24] Grønbech J, Sterndorff MJ, Grigorian H, Jacobsen V. Hydrodynamic modelling of wave-in-deck forces on offshore platform decks. In: Proceedings of the 33rd annual offshore technology conference 2001, Houston, Texas, USA, 2001, OTC 13189.
[25] Skjelbreia L, Hendrickson J. Fifth order gravity wave theory, in: Proceedings of seventh conference on coastal engineering, the Hague, the Netherlands, 1960. p. 184–196.
[26] Hansen K, Gudmestad OT, Reassessment of jacket type of platforms subject to wave-in-deck forces—current practice and future development. In: Proceedings of the 11th international offshore and polar engineering conference 2001. Stavanger, Norway; 2001.